

Indian Institute of Technology Madras
Department of Mathematics
MA2031-Linear Algebra for Engineers
Quiz-I

Time : 8.00 AM-8.50 AM

Date : Feb 20, 2023

Max Marks : 20

Instructions:

- (i) Answer **ALL** questions.
 - (ii) Write your name and **Roll number** on the answer book.
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1. Determine if the following sets are subspaces of $\mathbb{R}^3$. Justify your answers.

(a) $W = \{(x, y, z) \mid x^3 = y^3\}$. [3]

(b) $W = \{(x, y, z) \mid x^2 = y^2 + z^2\}$. [2]

2. Let $V = \text{span}\{(1, 0, 0), (1, -1, 0)\}$ and $W = \text{span}\{(0, 1, 0), (0, -1, 1)\}$.

(a) Find the dimensions of vector spaces $V + W$ and $V \cap W$. [3]

(b) Identify the vector space $V + W$. [1]

(c) Is $V + W$ a direct sum? [1]

3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$. Is the solution set W of the linear system $Ax = 0$ a subspace of $\mathbb{R}^3$?
Justify your answer. If yes, find a basis for W . [5]

4. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(x, y) = (x - y, 0, x + y)$. Let

$$B_1 = \{(1, 2), (1, 1)\} \text{ and } B_2 = \{(1, 1, 1), (1, 1, 0), (0, 1, 1)\}$$

be ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. Find the matrix representation $M_{B_1 B_2}(T)$ of the linear map T with respect to the bases B_1 and B_2 . [5]

1) (a) $W = \{(x, y, z) \mid x^3 = y^3\}$
 $= \{(x, y, z) \mid x = y\}$ —

1 mark

• Let (x, y, z) and $(a, b, c) \in W$.

$$(x, y, z) + (a, b, c) = (x+a, y+b, z+c).$$

Also $x+a = y+b$, since $x=y$ and $a=b$

Thus $(x, y, z) + (a, b, c) \in W$

1 mark

• Let $(x, y, z) \in W$ and $\alpha \in \mathbb{R}$.

Since $x=y$, $\alpha x = \alpha y$. Thus

$$\alpha(x, y, z) = (\alpha x, \alpha y, \alpha z) \in W.$$

1 mark

Thus W is a subspace of $\mathbb{R}^3$.

(b) $W = \{(x, y, z) \mid x^2 = y^2 + z^2\}$

Let For $(1, -1, 0)$ and $(1, 1, 0) \in W$, —

1 mark

$$(1, -1, 0) + (1, 1, 0) = (2, 0, 0) \notin W$$

1 mark

Thus W is NOT a subspace of $\mathbb{R}^3$

$$2) \quad V = \text{span} \{ (1, 0, 0), (1, -1, 0) \}, \quad W = \text{span} \{ (0, 1, 0), (0, -1, 1) \}$$

a) and (b)

Note: $V+W \supseteq \{ (1, 0, 0), (0, 1, 0), (0, -1, 1) \}$

Since $V+W$ is a subspace of $\mathbb{R}^3$,

$$V+W \supseteq \text{span} \{ (1, 0, 0), (0, 1, 0), (0, -1, 1) \} = \mathbb{R}^3$$

Thus $\mathbb{R}^3 \supseteq V+W \supseteq \mathbb{R}^3$. (ie) $V+W = \mathbb{R}^3$

Hence $\dim(V+W) = 3$

$$\begin{aligned} 3 = \dim(V+W) &= \dim(V) + \dim(W) - \dim(V \cap W) \\ &= 2 + 2 - \dim(V \cap W) \end{aligned}$$

$$\therefore \dim(V \cap W) = 1$$

Since $\dim(V \cap W) = 1$, $V \cap W \neq \{0\}$.

Thus $V+W$ is NOT a direct sum

Remark about Question-2

Wherever you claim something is a basis, or the span of something is $\mathbb{R}^3$ etc., without justification, you may lose a mark.

Q3 - Marking Scheme (Quiz - I, MA2031)

- If x_1 and x_2 are any two solutions of $Ax = 0$ then $A(x_1 + \lambda x_2) = Ax_1 + \lambda Ax_2 = 0 + 0 = 0$ (for any scalar λ). Therefore, the solution set $\{x\}$, of $Ax = 0$, is a subspace of R^3 **(2 marks)**
- Row transformations on the given system gives $x + 2y + 3z = 0$.
- One equation and Three unknowns, therefore, two unknowns can be fixed independently.
- Therefore, the dimension of the solution set is 2 **(1 mark)**
- Taking $(y \ z) = (0 \ 1)$ and similarly $(y \ z) = (1 \ 0)$ gives the basis as $\begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ **(2 marks)**

Question-4 (Marking Scheme)

$$\text{Let } B_1 = \{v_1 = (1, 2), v_2 = (1, 1)\}$$

$$B_2 = \{w_1 = (1, 1, 1), w_2 = (1, 1, 0), w_3 = (0, 1, 1)\}$$

$$T(v_1) = T(1, 2) = (-1, 0, 3) = a(1, 1, 1) + b(1, 1, 0) + c(0, 1, 1)$$

$$\Rightarrow \left. \begin{array}{l} a = 2 \\ b = -3 \\ c = 1 \end{array} \right\} \text{--- 2-marks}$$

$$T(v_2) = T(1, 1) = (0, 0, 2) = \alpha(1, 1, 1) + \beta(1, 1, 0) + \gamma(0, 1, 1)$$

$$\Rightarrow \left. \begin{array}{l} \alpha = 2 \\ \beta = -2 \\ \gamma = 0 \end{array} \right\} \text{--- 2-marks}$$

The matrix representation

$$M_{B_1 B_2}(T) = \begin{bmatrix} 2 & 2 \\ -3 & -2 \\ 1 & 0 \end{bmatrix} \text{--- 1-mark}$$